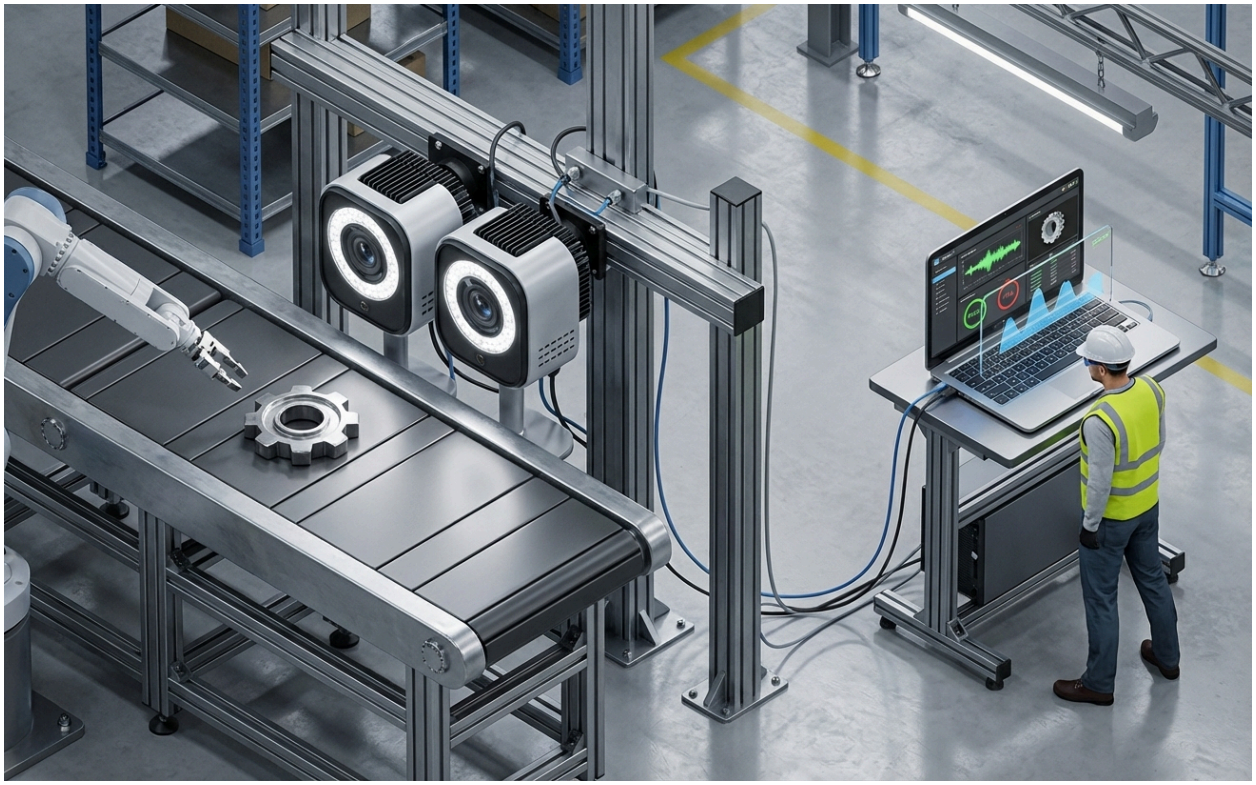
Home Task - AI-Guided Inspection System

Time: 1 Hour

The Challenge

Manufacturing parts pass through a multi-camera inspection station. On trigger from the controller, the system must capture frames from all cameras in parallel, run detection models per camera, then aggregate results into a unified report.

Your task: Create a one-shot prompt that guides an AI coding agent to build a project skeleton with mock implementations.



What You Deliver

1. **Your one-shot prompt** (`prompt.txt`)
2. **AI-generated project** that runs 10 mock inspection cycles
3. **Brief README** - your thinking, how to run, what you'd improve

The System Flow

Trigger → Parallel Capture (2 cams) → Per-Camera [Preprocess → Inference → Post-process] → Aggregate Results → Report

Evaluation (120pts)

- System Design (20pts)
- Product Mindset (20pts)
- Time Management & Mitigations (20pts)
- Project Structure (20pts)
- Performance & Stability Mindset (20pts)
- **Bonus**: Runnable demo (20pts)

What We're Testing

Can you translate business context into technical architecture that an AI can execute? Your prompt must guide the AI to address:

- Parallel execution design
- Proper abstractions and separation of concerns
- Manufacturing context and real-world constraints
- Error handling and observability
- Scalability considerations

Rules

- **Mock everything** - no real image processing or ML models
- Focus on architecture, not algorithms
- Show the flow, not the implementation details
- Structure for production mindset

The quality of your prompt determines the quality of the output. Think carefully about what an AI needs to build a well-architected system.